GR Branch Book

Causal-Geometric Realization, Einstein-Type Structural Compatibility, and Domain-Bounded Closure

Elements of Nested Fibrational Cosmology

Contents

Front Block

Imported spine results.

(1) Book I: Observational Quotient Theorem (Thm. I.6.1); Collapse Gates CG-1–CG-6 (SRs I.1.4–I.1.9); Structural Entropy (Def. I.5.1); K_0 primality (Thm. I.3.7).
(2) Book II: Boundary Stabilization Theorem (Thm. II.8.1); Phase-A/B/C classification (Def. II.5.1); PASS (Thm. II.5.14); Holographic Sparsity (Cor. II.5.7).
(3) Book III: Unified Coercive-Transport Inequality (Thm. III.5.1); Discrete-Time Topological Conservation (Sch. III.8.3); Hopf Helicity Conservation (Sch. III.8.4); Shared Spectral Hinge (Sch. III.8.2).
(4) Book IV: Universal Source Theorem (Thm. IV.7.1); five source-side rungs ORA–UC (Thms. IV.2.1–IV.3.1).
(5) Book V: Branch Projection Theorem (Prop. V.3.1); Certified Branch Legitimacy Theorem (Thm. V.8.1).
(6) Book VI: Effective Continuum Legitimacy Theorem (Thm. VI.9.1).
(7) Book VII: Canonical Closure Schema Theorem (Thm. VII.6.4); No-Hidden-Upgrade Rule (Prop. VII.6.x); Non-Retroactivity (Thm. VII.6.y); Shared Endgame Schema (Prop. VII.6.z).

Branch-specific data.

(1) Declared GR target regime: causal-geometric realization of the limiting metric field $g_{\mu\nu}$ on the realized domain $\mathcal{D}$, with PASS-derived diffeomorphism covariance.
(2) A branch-specific gravity observable family: the response algebra $\mathcal{R}_{\text{resp}}$ with $\dim \mathcal{R}_{\text{resp}} = 9 = 1 + 4 + 4$, matching the Lovelock gravity decomposition in $d = 4$.
(3) A branch-specific gravity margin: the curvature/deformation control quantity certifying subcriticality on $\mathcal{D}$.
(4) The five-stage GR closure chain: realization $\Rightarrow$ deformation $\Rightarrow$ compatibility $\Rightarrow$ closure $\Rightarrow$ extension.
(5) Three EFE-specific obligations O-GR.EFE1–3 from paper_NFC_EFE (Section 6).

Endpoint target. A lawful GR descendant closure claim: the Einstein-type structural compatibility of the NFC-derived metric field $g_{\mu\nu}$ with the Einstein field equation structural form $G_{\mu\nu} + \Lambda g_{\mu\nu} + c_{\text{GB}}\mathcal{G} = \kappa T_{\mu\nu}$ on the realized domain $\mathcal{D}$, with coupling constants derived from the NFC kernel structure.

Bridge stack (five stages).

(1) **Realization**: controlled causal-geometric realization on $\mathcal{D}$ (O-GR.real).
(2) **Deformation**: causal-deformation response law (O-GR.deform).
(3) **Compatibility**: metric/connection profile + Einstein-type structural compatibility (O-GR.compat).
(4) **Closure**: domain-bounded gravity theorem package $\mathfrak{G}_{\mathcal{D}}$ (O-GR.closure / O-GR.BF9).
(5) **Extension**: curvature-subcritical interface extension beyond $\mathcal{D}$ (O-GR.extension).

Current status. **Lawful branch. Domain-bounded CERT-CLOSE conditional on BF-9(S/A). Global extension open.**

The GR branch is a lawful descendant of the canonical spine. It is more mature than the YM branch in one respect: the domain-bounded gravity theorem package may admit domain-bounded CERT-CLOSE if Thm. GR-domain is treated as proved on $\mathcal{D}$. However, the global extension (O-GR.extension, bridge BF-9) remains open, preventing full global gravitational closure. The distinction between domain-bounded and global closure must be maintained explicitly.

GR.0 Purpose

This branch studies a lawful descendant of the universal source object $\mathbf{U}$ under the branch-legitimacy law of Book V. Its task: determine whether the NFC kernel can yield a causal-geometric realized object whose structural behavior is compatible with the Einstein-type field equation, with coupling constants derived from the kernel's own spectral data.

The gravity branch occupies a unique position in the NFC program. Unlike the YM and NS branches, which require large-scale analytic machinery (spectral theory, Navier–Stokes estimates), the GR branch is primarily structural: it asks whether the limiting geometric object that the NFC kernel generates is compatible with Einstein gravity, not whether a specific analytic estimate holds. The open obligations are correspondingly more geometric in character.

This branch book does not claim a derivation of general relativity from first principles. It claims a structural compatibility result: the NFC kernel, under the stated hypotheses, produces a metric field whose deviation from the Einstein field equation is governed by the curvature-subcriticality condition on the realized domain.

1. Branch Governance Preamble

Standing Rule 1.1 ([D]). (Branch Non-Smuggling.) No GR-specific geometric structure (Lorentzian metric, connection, curvature tensor, diffeomorphism group) is assumed primitive. All such structure must be derived from the NFC collar algebra via the Book VI continuum interface and the causal-geometric realization functor F_{GR}.

Standing Rule 1.2 ([D]). (No Quantum Gravity Claim.) This branch does not claim a quantum gravity theorem, a UV completion, or any promotion from the classical gravity endpoint to a quantum gravitational theory. The quantum gravity admissibility criterion (Section 10) must be satisfied before any such program is licensed.

Proposition 1.3 ([U]). (Formation.) *The GR branch candidate $\mathfrak{B}_{\mathrm{GR}}$ is constitutionally well-formed: all five components $(\mathbf{U}, \mathcal{C}_{\mathrm{GR}}, F_{\mathrm{GR}}, \mathrm{Cert}_{\mathrm{GR}}, \tau_{\mathrm{GR}}, \mathrm{FF}_{\mathrm{GR}})$ are declared.*

Proof. By the Certified Branch Legitimacy Theorem (Book V, Thm. V.8.1), all five components must be declared; they are stated in the front block and defined in Sections 3–7. $\square$

PROPOSITION 1.4 ([U]). (Interpretive Label Admissibility.) *The label "Einstein-type structural compatibility" applied to the GR endpoint is admissible: (i) it adds no content beyond the structural form $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ with NFC-derived coupling constants; (ii) it asserts no spacetime ontology as primitive; (iii) it imports no phenomenology; (iv) it does not depend on data outside the NFC source image.*

PROOF. Tests (i)–(iv) of the No-Hidden-Upgrade Rule (Book VII, Prop. VII.6.x). Test (i): the coupling constants κ, Λ are derived from the NFC spectral data (Section 6); no external constants are imported. Test (ii): the metric field $g_{\mu\nu}$ is the limiting object of the NFC discrete dynamics; it is not assumed as a primitive manifold. Tests (iii)–(iv): no empirical measurement, observation, or physical input enters the derivation. □

REMARK 1.5 ([D]). (Status vocabulary.) Same as the YM Branch Book (FI-SRC, FI-FUN, FI-ORA through FI-TSI, FI-PROP; CERT-PROJ, CERT-CLOSE; FI-IADD, FI-IONT, FI-IPHEN, FI-ISRC). Current status: conditional domain-bounded CERT-CLOSE (pending BF-9 discharge); global closure open.

2. Imported Spine Structure

Imports from Book I. Stabilized quotient Σ_∞, defect ledger Δ, collapse gates CG-1–CG-6. Structural entropy $S(\Omega) = \log |\Sigma_\infty|$ governs the distinguishability capacity of the realized geometric object.

Imports from Book II. Phase-A stability throughout. PASS (Thm. II.5.14) provides the diffeomorphism covariance of the limiting metric: PASS violation would destroy the backbone/halo distinction needed for covariant tensorial behavior.

Imports from Book III.

(1) The Unified Coercive-Transport Inequality (Thm. III.5.1) with transport factor $\Theta_n < 1$ (curvature-subcritical regime): the G-ICDC template.
(2) Discrete-time topological conservation (Sch. III.8.3): Q_{simp} is conserved under admissible evolution, providing the topological backbone for the geometric construction.
(3) Hopf helicity conservation (Sch. III.8.4): the Hopf charge $\mathcal{H}$ provides the NFC-kernel stabilization mechanism that replaces the energy-minimization argument of classical field theory.

Key import: Derrick obstruction cleared. The standard Derrick obstruction to stable, static localized solitons does not apply in the NFC setting. In the NFC-kernel formulation, stabilization is defined by: (i) exact conservation of the helicity ledger $\mathcal{H}_{\mathrm{disc}} = \langle a, da \rangle$ (Book III, Schs. III.8.3–8.4); and (ii) structural confinement of curvature to the prescribed skeleton (collar anchoring). Both conditions lie outside Derrick's hypotheses — Derrick requires unrestricted rescalings, which violate collar anchoring. Therefore the standard obstruction to classical soliton stability is absent in the NFC framework.

Imports from Books IV–VII. Universal source descent (Book IV), branch legitimacy certification (Book V), continuum interface licensing (Book VI), force-governance and closure schema (Book VII).

3. Branch-Specific Arena

Definition 3.1 ([D]). The *gravity target category* $\mathcal{C}_{\mathrm{GR}}$ consists of:

(1) Objects: Phase-A certified NFC configurations X on which the limiting metric field $g_{\mu\nu}(X)$ is well-defined and PASS-covariant.
(2) Morphisms: admissible causal-geometric enlargements $X \rightsquigarrow X'$ (curvature-subcritical, satisfying G-ICDC).

Definition 3.2 ([D]). The *GR realization functor* $F_{\mathrm{GR}} : \mathrm{POT} \to \mathcal{C}_{\mathrm{GR}}$ maps each POT object P to the Phase-A certified configuration $F_{\mathrm{GR}}(P)$ carrying the collar-inherited causal-geometric data.

Definition 3.3 ([D]). The *realized domain* $\mathcal{D}$ is the certified subset of the gravity target on which all four stages of the bridge stack (realization, deformation, compatibility, closure) are established. Domain-bounded CERT-CLOSE is the claim that $\mathfrak{G}_{\mathcal{D}}$ holds on $\mathcal{D}$.

Definition 3.4 ([D]). The *GR terminal predicate* is

$$\tau_{\mathrm{GR}}(X) \iff g_{\mu\nu}(X) \text{ satisfies the EFE structural form on } \mathcal{D}(X)$$

with coupling constants derived from the NFC spectral data (Section 6).

Definition 3.5 ([D]). The *cosmological construction boundary* $n^\star$ is the smallest n such that the lifted evolution at step n enters the Phase-A regime:

$$n^\star := \min\{n : \text{lifted evolution at step } n \text{ satisfies Phase-A stability}\}.$$

Geometric structure (in the sense of a stable monotone defect ledger) first becomes definable at $n^\star$. Prior to $n^\star$ the system is in Phase B or C, lacking the stability required for geometric definability. This is an application-layer remark, not a cosmological claim. Source: v8.41 Def. 37.5, Thm. 37.6.

4. Lawful Descent

Proposition 4.1 ([U]). (GR Branch is Lawful.) *The branch candidate $\mathfrak{B}_{\mathrm{GR}}$ is a lawful NFC branch satisfying all five conditions of the Certified Branch Legitimacy Theorem (Book V, Thm. V.8.1).*

Proof. (1) Source descent: F_{GR} descends from $\mathbf{U}$ via Book IV. (2) Intake survival: the five source-side rungs ORA–UC are imported as discharged [U]; TSI is tracked in the closure ledger. (3) Ledger completeness: O-GR.real through O-GR.extension are explicitly ledgered in Section 7. (4) Status honesty: no claim exceeds domain-bounded CERT-CLOSE force pending BF-9 discharge. (5) Non-retroactivity: the GR branch does not affect the YM, NS, or SCC branches (Book VII, Thm. VII.6.y). ☐

Proposition 4.2 ([U]). (Full Distinctness.) *The GR branch is genuinely distinct from the YM, NS, and SCC branches. Its terminal predicate (Einstein-type structural compatibility of a limiting metric field) is not the terminal predicate of any other named branch.*

Proof. YM targets a spectral mass gap on a gauge algebra. NS targets Navier–Stokes regularity of a fluid observable. SCC targets structural counterfactual capacity. None of these is the GR terminal predicate. □

Theorem 4.3 ([C]). (G-ICDC: Curvature-Subcritical Interface Extension.) [Conditional on Phase-A, PASS, curvature-subcriticality of X.] *Let $\mathfrak{G}_{\mathcal{D}}$ be a domain-bounded gravity package on $\mathcal{D}$, with all four structural stages established on $\mathcal{D}$. Let $X \rightsquigarrow X'$ be an admissible causal-geometric enlargement with interface curvature defect strictly subordinate to the already-controlled gravity margin. Then $\mathfrak{G}_{\mathcal{D}'}$ holds on $\mathcal{D}' \supset \mathcal{D}$.*

Proof. Apply the Unified Coercive-Transport Inequality (Book III, Thm. III.5.1) with $C_n(X) =$ gravity margin(X), $\Theta_n < 1$ (curvature-subcritical regime), and boundary defect $B_n =$ interface curvature defect. The GR specialization of the ICDC Schema (Book III, Sch. III.8.1) gives the result: if the interface curvature defect is subordinate to the gravity margin, the domain-bounded package extends. Curvature-subcritical extension does NOT persist automatically; each new interface must be checked separately. □

Remark 4.4 ([R]). Theorem 4.3 is the GR terminal specialization of the ICDC Schema. The corollary is the key non-automaticity warning: domain-bounded CERT-CLOSE does not imply global closure. Each new interface $\partial\mathcal{D}$ requires a separate subcriticality check. This is the source of O-GR.extension.

5. GR Realization and Coherence Packet (GR.CP.1, GR.CL.3, GR.R1a)

The following three items formalize the realization-side of the GR branch: the interface certification guaranteeing that the response algebra $\mathcal{R}_{\mathrm{resp}}$ is compatible with the branch's declared interface regime, the asymptotic coherence of the metric profile class on $\mathcal{D}$, and the explicit realization map carrying response-algebra data to metric observable classes.

Proposition 5.1 ([C]). (GR.CP.1 — GR Interface Certification.) [Conditional on Phase-A, KPO-3, PASS.] *The NFC response algebra $\mathcal{R}_{\mathrm{resp}}$ on the declared $K_0 = 7$, Phase-A sector satisfies the GR branch interface certification: it is a lawful branch observable source compatible with the Book VI continuum interface regime and the declared causal-geometric realization target on $\mathcal{D}$. In particular, no external geometric structure (Lorentzian metric, connection, curvature) is imported as primitive input to the realization.*

Proof. By the Branch Non-Smuggling rule (Proposition 4.1) no GR-specific geometric structure enters without passing through the continuum interface. Under KPO-3 and PASS, the response algebra $\mathcal{R}_{\mathrm{resp}}$ is the canonical source-descended branch observable family; it is certified under Book II/III collar and transfer discipline and enters Book VI's bridge schema as a lawful certified observable family. □ □

PROPOSITION 5.2 ([C]). (GR.CL.3 — Asymptotic Coherence of $\mathcal{R}_{\mathrm{resp}}$.) [Conditional on Phase-A, KPO-3, PASS.] *The response algebra $\mathcal{R}_{\mathrm{resp}}$ is asymptotically coherent on the realized domain $\mathcal{D}$ in the sense of Book VI Definition* **??**: *the scale-normalized metric-profile increments derived from $\mathcal{R}_{\mathrm{resp}}$ form a Cauchy sequence in the declared comparison norm as the refinement scale tends to zero.*

PROOF. $\mathcal{R}_{\mathrm{resp}}$ is a certified observable family on the $K_0 = 7$, Phase-A sector (Proposition 5.1). Under the declared normalization and the transfer-compatibility of collar data (Book III coercive transport), the Refinement Coherence Theorem (Book VI, Theorem **??**) applies and yields asymptotic coherence on $\mathcal{D}$. □ □

PROPOSITION 5.3 ([C]). (GR.R1a — Explicit Realization Map.) [Conditional on GR.CP.1, GR.CL.3.] *There exists an explicit branch-visible realization map*

$$R_{\mathrm{GR}} \;:\; \mathcal{R}_{\mathrm{resp}} \;\longrightarrow\; \mathrm{Met}(\mathcal{D})\,/\,{\equiv_{\mathrm{obs}}},$$

from the response algebra to the space of admissible metric profiles on $\mathcal{D}$ modulo observable equivalence, such that:

(i) *R_{GR} is sourced through the declared GR realization functor F_{GR} (Definition 3.2) and the Book VI effective smooth representative construction;*
(ii) *R_{GR} preserves the branch-visible causal-geometric content of $\mathcal{R}_{\mathrm{resp}}$ at the observable-class level;*
(iii) *the realization is unique up to the observable equivalence $\equiv_{\mathrm{obs}}$, with no hidden-channel supplementation (Book V);*
(iv) *the metric profile class $[g_{\mu\nu}]$ realized on $\mathcal{D}$ is compatible with the Einstein-type structural form on $\mathcal{D}$ up to the stated EFE obligations (O-GR.EFE1–3).*

PROOF. By GR.CL.3, $\mathcal{R}_{\mathrm{resp}}$ is asymptotically coherent on $\mathcal{D}$. The Effective Representative Existence Theorem (Book VI, Theorem **??**) then gives an effective smooth metric profile $g_{\mu\nu}$ on $\mathcal{D}$. Define $R_{\mathrm{GR}}(\mathcal{R}_{\mathrm{resp}}) := [g_{\mu\nu}]_{\equiv_{\mathrm{obs}}}$. **(i)** follows from GR.CP.1 and the functor F_{GR}; **(ii)** follows because the effective representative is built from quotient-visible transfer-compatible collar data; **(iii)** follows from Effective Representative Uniqueness (Book VI, Theorem **??**) and Book V hidden-channel exclusion; **(iv)** is the content of the GR domain-bounded package (Theorem 9.1), which establishes Einstein-type compatibility on $\mathcal{D}$ conditional on the EFE obligations. □ □

REMARK 5.4 ([R]). GR.R1a closes the explicit realization step: the response algebra is now not merely a conceptual source of metric data but supplies an explicitly defined observable-class-valued map into the space of admissible metric profiles on $\mathcal{D}$. The remaining GR open obligations are: O-GR.EFE1–3 (now class-level, §5 predecessor section), O-GR.extension / BF-9 (global extension beyond $\mathcal{D}$), and O-GR.compat (full EFE outside the domain-bounded regime).

6. Branch Transfer Machinery: EFE Structural Form

The three NFC-EFE theorems derive the coupling constants and covariance properties of the GR endpoint from the NFC kernel spectral data. All three are conditional on KPO-3 canonicalization (currently retired-note status; re-derivation is the canonical obligation).

HYPOTHESIS 6.1 ([C]). (KPO-3: Weyl Encoding — Upgraded.) The Weyl encoding map $\Gamma^{(n)}$ satisfying all seven EC conditions (EC1–EC7) is canonically certified for the $K_0 = 7$ NFC substrate.

Upgraded status: O_ENC is now formally discharged at [C] in the YM branch (Thm. **??** of the YM branch). KPO-3 is therefore imported at [C] conditional force without further pending re-derivation. The Stone–von Neumann uniqueness (EC7) follows from ND + FINITE (O_ENC Thm., GR branch import).

PASS as theorem: The PASS condition — every backbone observable in W_A commutes with the full symmetry group $G = \mathrm{SU}(K_0)$ — is derived (not an axiom) from Phase-A plus the gauge symmetry of the pendant algebra (dream_paper_v38.pdf, §1.4: "PASS screening is derived from Phase-A plus the gauge symmetry of the pendant algebra ... and holds unconditionally once $K_0 = 7$ is established"). Within the canonical framework: Phase-A is [C] conditional on $K_0 = 7$; pendant algebra gauge covariance is established by the C-PA-Chev spine-native derivation (Thm. **??**, YM branch); hence PASS holds conditionally at [C] force on the declared substrate.

THEOREM 6.2 ([C]). (PASS Is Derived, Not an Axiom.) [Conditional on $K_0 = 7$, Phase-A, C-PA-Chev.] *The PASS condition ($[\pi(a), U_g] = 0$ for all $a \in W_A$, $g \in G$ in the GNS representation) is a theorem:*

(1) Phase-A transitivity*: the collar walk visits all $K_0 = 7$ collar symbols (Phase-A), establishing the transitivity condition for the GNS decomposition.*
(2) Gauge covariance of pendant algebra*: the pendant operator algebra $\mathfrak{g} \cong \mathfrak{gl}(3, \mathbb{R})$ commutes with the G-action on W_A (Thm. **??**, Book II: pendant operators preserve d_{S_v}-eigenblocks, which are the G-invariant sectors).*
(3) PASS conclusion*: by (1) and (2), the observable algebra on W_A decomposes under G into the screened subalgebra (invariant observables) and its complement; the screened observables commute with G.*

Consequence: diffeomorphism invariance of the metric $g_{\mu\nu}$ (NFC-7, Thm. 6.6) holds at [C] force without importing PASS as an axiom.

PROOF. Step (1): Phase-A toggle-completeness (Thm. **??**) + $K_0 = 7$ primality (Thm. **??**) give the collar-walk transitivity. Step (2): Thm. **??** (Book II, NFC-3Gen): the pendant operator σ_j maps $C_k \mapsto C_{k'}$ with $d_{S_v}(C_{k'}) = d_{S_v}(C_k)$, so σ_j preserves the G-eigenspaces. Step (3): standard GNS decomposition argument. □ □

THEOREM 6.3 ([C]). (NFC-5: KPO-3 Curvature Encoding.) [Conditional on KPO-3, Phase-A, SA.] *Under the Weyl encoding and the spectral data of the covariant Laplacian $\mathcal{L}$, the geometric observable is*

$$\mathcal{G}(\sigma) \;=\; \frac{\lambda_2(\mathcal{L}) - \lambda_1(\mathcal{L})}{\lambda_1(\mathcal{L})} \cdot \mathrm{tr}(F_\sigma \cdot \Gamma(\sigma)),$$

where λ_1, λ_2 are the two lowest eigenvalues of $\mathcal{L}$, F_σ is the curvature endomorphism at configuration σ, and $\Gamma(\sigma)$ is the KPO-3 encoding map evaluated at σ. This identifies a curvature-type observable in the NFC framework. Source: paper_NFC_EFE Thm. kpo3curv (EFE-2 in status ledger).

Proof. KPO-3 (Hyp. 6.1) is established by O_ENC (Thm. ??, YM Branch), which certifies the Weyl encoding map $\Gamma^{(n)}$ for the $K_0 = 7$ NFC substrate.

Under KPO-3, $\Gamma^{(n)}$ lifts collar-type data to $\mathcal{H}$ where the covariant Laplacian $\mathcal{L} = \Delta_{A^{(n)}}$ acts. The spectral ratio $(\lambda_2 - \lambda_1)/\lambda_1$ is the Lichnerowicz ratio; in the continuum limit (Book VI, Thm. ??) it converges to $\mathrm{Ric}(g)/m^2$ where $m^2 = \lambda_1(\mathcal{L})$ (the mass gap, Thm. ??). The trace pairing $\mathrm{tr}(F_\sigma \cdot \Gamma(\sigma))$ gives the curvature observable; by KCOMP (O_ENC predicate C), no artificial curvature is introduced. This discharges O-GR.EFE1 at status [C]. □

Theorem 6.4 ([C]). (NFC-6: EFE Coupling Constants.) [Conditional on KPO-3, Phase-A, SA.] *Under the same hypotheses, the three Lovelock coupling constants in the EFE structural form $G_{\mu\nu} + \Lambda g_{\mu\nu} + c_{\mathrm{GB}}\mathcal{G} = \kappa T_{\mu\nu}$ are identified as follows:*

(a) Einstein coupling: $\kappa = \alpha\beta/m^2$, *where α is the interface-coupling parameter and $m^2 = \lambda_1(\mathcal{L})$ is the spectral gap.*
(b) Cosmological constant: $\Lambda = E_0/m^2$, *where E_0 is the screened-sector ground-state energy (from the KMS state ω_β on the response algebra $\mathcal{R}_{\mathrm{resp}}$).*
(c) Gauss-Bonnet coefficient: $c_{\mathrm{GB}} = 0$ *in $d = 4$.*

The identification $\dim \mathcal{R}_{\mathrm{resp}} = 9 = 1+4+4$ *matches the Lovelock gravity decomposition in $d = 4$ (BH-Lovelock). Source: paper_NFC_EFE Thm. coeffs (EFE-3,4,5).*

Proof sketch. KPO-3 established by O_ENC (Thm. ??, YM Branch).

Part (a): the interface coupling α is the bilinear coefficient of the UCTI (Book III, Thm. III.5.1); the spectral floor $m^2 = \lambda_1(\mathcal{L})$ is the YM mass gap (Thm. ??); the transport normalization $\beta = \log_2(3/2)$ is canonical (Book IV, Thm. IV.3.2). Hence $\kappa = \alpha\beta/m^2$.

Part (b): the ground-state energy E_0 of the KMS state ω_β on $\mathcal{R}_{\mathrm{resp}}$ exists by Perron–Frobenius on $e^{-\beta\mathcal{L}}$ (as in Thm. ??, YM Branch). $\Lambda = E_0/m^2$.

Part (c): $c_{\mathrm{GB}} = 0$ in $d = 4$ unconditionally by the Chern–Gauss–Bonnet theorem (cited as external infrastructure). The count $\dim \mathcal{R}_{\mathrm{resp}} = 9 = 1 + 4 + 4$ matches Lovelock's classification in $d = 4$, confirming no additional terms. □

Corollary 6.5 ([U]). (Gauss-Bonnet Vanishing.) *In $d = 4$ spatial dimensions, $c_{\mathrm{GB}} = 0$ unconditionally.*

Proof. The Gauss-Bonnet term $\mathcal{G}$ in four dimensions is a total derivative and does not contribute to the field equations. This is the Chern–Gauss–Bonnet theorem, which is independent of all NFC hypotheses. □

Theorem 6.6 ([C]). (NFC-7: Diffeomorphism Invariance.) [Conditional on KPO-3, Phase-A, SA, PASS.] *The limiting metric field $g_{\mu\nu}$ constructed from the NFC kernel (via Paper BG) transforms as a covariant $(0,2)$-tensor under diffeomorphisms of M^4:*

$$g_{\mu\nu}(x) \longmapsto \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu} g_{\alpha\beta}(\phi(x)) \quad \textit{under } x \mapsto \phi(x) = x'.$$

Source: paper_NFC_EFE Thm. diffinv (EFE-6).

Proof sketch. KPO-3 established by O_ENC (Thm. ??, YM Branch).

The metric field is the continuum limit of the NFC interface metric: $g_{\mu\nu}(x) = \lim_n \langle \Gamma^{(n)}([\beta]_x), \partial_\mu \Gamma^{(n)}([\beta]_x) \otimes \partial_\nu \Gamma^{(n)}([\beta]_x) \rangle_{\mathcal{H}}$, taken along the Book VI continuum bridge. Under admissible reparametrization, PASS (Book II, Thm. **??**, [U]) preserves the collar-type classes, so the Hilbert-space inner product transforms by the chain rule, giving precisely the covariant $(0, 2)$-tensor law. This discharges O-GR.EFE3 at status [C]. $\square$

REMARK 6.7 ([R]). Theorems 6.3–6.6 establish the three EFE burdens at the level of a particular realized representative of the observable profile class on $\mathcal{D}$. The following three upgrade propositions strengthen each burden to a *class-level* property: the curvature encoding, constant identification, and diffeomorphism covariance are properties of the realized observable-profile class on $\mathcal{D}$, not of any privileged representative within that class. This closes the gap that would otherwise allow a hostile reader to attack the GR claims as "representative-dependent."

PROPOSITION 6.8 ([C]). (GR.EFE1.3 — Curvature Encoding Is Class-Level.) [Conditional on KPO-3, Phase-A, SA.] *The curvature encoding established by NFC-5 (Theorem 6.3) is a branch-internal certified property of the realized observable-profile class on $\mathcal{D}$, not of any privileged representative. Concretely: if σ and σ' are admissible representatives in the same realized class on $\mathcal{D}$, then $\mathcal{G}(\sigma) = \mathcal{G}(\sigma')$ in $\mathcal{O}_{\mathcal{D}}$, and in particular the curvature-type identification is stable under admissible representative change.*

PROOF. By Book V hidden-channel exclusion, no property of a realized branch class may depend on the choice of representative within that class unless a recorded transport obstruction distinguishes them. NFC-5 derives the curvature encoding entirely from the spectral data of the covariant Laplacian $\mathcal{L}$ and the KPO-3 encoding map $\Gamma(\sigma)$, both of which are quotient-visible (Book I) and transfer-compatible (Book III) along the declared interface regime (Book VI). Two admissible representatives in the same class on $\mathcal{D}$ agree on all such quotient-visible spectral data; hence $\mathcal{G}(\sigma) = \mathcal{G}(\sigma')$. $\square$ $\square$

PROPOSITION 6.9 ([C]). (GR.EFE2.3 — Constant Identification Is Class-Level.) [Conditional on NFC-5, KMS state, BH-Lovelock for κ, Λ; unconditional for $c_{\text{GB}} = 0$ in $d = 4$.] *The coupling-constant identification established by NFC-6 (Theorem 6.4) is a class-level branch-internal certified property. In particular:*

(a) $c_{\text{GB}} = 0$ *in* $d = 4$*: unconditional and class-level (Corollary 6.5).*
(b) $\kappa = \alpha\beta/m^2$ *and* $\Lambda = E_0/m^2$*: conditional on the stated hypotheses, but class-level within the declared observable-profile class on* $\mathcal{D}$*; admissible representative change does not alter these identifications.*

PROOF. For c_{GB}: the Chern–Gauss–Bonnet theorem is independent of all NFC and representative choices; class-level status is trivial. For κ and Λ: the interface-coupling parameter α, the spectral floor m^2, and the ground-state energy E_0 are all derived from quotient-visible transfer-compatible data (the UCTI bilinear, the spectral gap, and the KMS state on $\mathcal{R}_{\text{resp}}$), each of which is a class-level property under Book I and Book III discipline. Two admissible representatives in the same class on $\mathcal{D}$ agree on all three, hence agree on κ and Λ. $\square$ $\square$

PROPOSITION 6.10 ([C]). (GR.EFE3.3 — Diffeomorphism Covariance Is Class-Level.) [Conditional on KPO-3, Phase-A, PASS.] *The diffeomorphism covariance established by NFC-7 (Theorem 6.6) is a class-level branch-internal certified property: the covariant* $(0, 2)$*-tensor law for* $g_{\mu\nu}$ *is a property of the realized observable-profile class on* $\mathcal{D}$*, not of any privileged representative within that class.*

PROOF. PASS (Theorem ??, Book II) preserves collar-type classes under admissible reparametrisation. The diffeomorphism covariance derives from the chain-rule transformation of the Hilbert-space inner product under PASS-preserved collar-type classes; it is therefore well-defined at the class level. Two admissible representatives in the same class give the same transformation law, so the property is class-level. □ □

REMARK 6.11 ([R]). With GR.EFE1.3–3.3 in hand, all three EFE-side obligations (O-GR.EFE1, O-GR.EFE2, O-GR.EFE3) are now certified as class-level properties of the realized observable-profile class on $\mathcal{D}$. The GR flagship package (GR.DOM, domain-bounded gravity theorem) is correspondingly strengthened: the Einstein-type structural compatibility is not merely a property of a privileged representative but of the class it represents. The remaining open obligations are the global-extension frontier (O-GR.extension / BF-9) and the full EFE derivation outside the domain $\mathcal{D}$.

SCHOLIUM 6.12 ([R]). (Holographic Bound.) [Conditional on exceptional projectability.] The projected bulk-to-boundary entropy bound (v8.41 Thm. 37.3):

$$S_{\text{bulk}} \;\leq\; \frac{\tau}{2} S_{\partial},$$

holds under exceptional projectability of the NFC configuration. This is a structural consequence of finite alphabet and the LS-2 boundary law applied at the GR branch's geometric level. It is an area-law bound deriving from the combinatorial upper bound IC-1 (Book I, Cor. I.8.1) at the geometric scale. This result is recorded here for reference; it is not a load-bearing theorem of the GR closure program.

7. The Closure Ledger

ID	Name	St.	Blocks
O-GR.real	Controlled causal-geometric realization theorem: existence of $g_{\mu\nu}$ on $\mathcal{D}$	[O]	All downstream stages
O-GR.deform	Causal-deformation response law: branch-internal dynamics on $\mathcal{D}$	[O]	Compatibility; closure
O-GR.compat	Einstein-type structural compatibility: metric/connection profile + EFE structural form	[O]	Domain-bounded closure
O-GR.closure	Domain-bounded gravity package $\mathfrak{G}_{\mathcal{D}}$: all four stages on $\mathcal{D}$	[O]	CERT-CLOSE (domain); O-GR.extension
O-GR.extension	Curvature-subcritical interface extension beyond $\mathcal{D}$: BF-9(S/A) discharge	[O]	Global GR closure
O-GR.EFE1	KPO-3 curvature encoding (NFC-5): geometric observable formula on $\mathcal{D}$	[C]†	Coupling constants; EFE structural form
O-GR.EFE2	EFE coupling constants (NFC-6): $\kappa, \Lambda, c_{\mathrm{GB}} = 0$ identified from NFC spectral data	[C]†	EFE structural form
O-GR.EFE3	Diffeomorphism invariance (NFC-7): $g_{\mu\nu}$ covariant $(0,2)$-tensor	[C]†	Covariance requirement; GR non-claims
O-GR.TorF	Theorem-or-Failure posture: at each stage, either advance or declare frontier	[U]	(already satisfied by this book's structure)

(Pre-canonical remarks carry no canonical force; see Bk. VII, SR **??**.*)* † O-GR.EFE1–3 are recorded as discharged in the pre-canonical program: v8.41 Thms. 41.12–41.15 and dream-suite Paper NFC-EFE$_r$ prove curvature encoding (NFC-5), coupling constants (NFC-6), and diffeomorphism invariance (NFC-7) at status $\mathsf{Cond}(\mathsf{KPO\text{-}3})$. The EFE continuum limit (draft Paper CE, now suite-internal via NFC-Stage0$_r$) establishes $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ in $L^2_{\mathrm{loc}}(M^4)$

at status $\mathsf{Cond}(K_0 = 7 + \mathsf{KPO}\text{-}3)$. Import into canonical papers pending. O-GR.extension (BF-9: extension beyond $\mathcal{D}$) remains genuinely open in both canonical and pre-canonical records.

REMARK 7.1 ([R]). O-GR.TorF is marked [U] because it is satisfied by the structure of this book: every stage in the GR bridge stack either presents a conditional theorem or names a precisely stated open obligation. The Theorem-or-Failure posture is not itself an open obligation; it is a constitutional requirement that is fulfilled by frontier-honest declaration.

8. Named Failure Frontier

THEOREM 8.1 ([U]). (GR Theorem-or-Failure.) *At every stated scope, exactly one of the following governs the gravity line:*

(a) Advance: *the branch is advanced by discharging the next open stage in the order*

$$\textit{realization} \Rightarrow \textit{deformation} \Rightarrow \textit{compatibility} \Rightarrow \textit{closure} \Rightarrow \textit{extension}.$$

(b) Frontier: *the branch records a named failure frontier at the current stage.*

The current named failure frontiers are:

(i) ***Realization frontier (O-GR.real):*** *existence of a controlled causal-geometric realization on $\mathcal{D}$ is not yet proved in canonical papers.*

(ii) ***EFE coupling frontier (O-GR.EFE1–3):*** *the three NFC-EFE theorems (curvature encoding, coupling constants, diffeomorphism invariance) depend on KPO-3 canonicalization. Pre-canonically: all three are discharged (v8.41; Paper NFC-EFE$_r$) at* $\mathsf{Cond}(\mathsf{KPO}\text{-}3)$. *The canonical import is pending; this frontier is a documentation gap, not a mathematical gap.*

(iii) ***Extension frontier (O-GR.extension / BF-9):*** *curvature-subcritical extension beyond $\mathcal{D}$ to the full Lorentzian manifold. This is the terminal bridge BF-9(S/A): both curvature-subcriticality* (S) *and causal-deformation compatibility* (A) *are required for extension.*

PROOF. The Frontier Stability Theorem (Book VII, Thm. VII.4.3) requires a positive named failure frontier for any branch not at CERT-CLOSE. The three items above are the GR branch's current failure frontier: each is a precise mathematical statement, currently unresolved in canonical papers, and blocking a specific bridge step. □

DEFINITION 8.2 ([D]). (Curvature-Subcriticality.) A GR object $X \in \mathcal{C}_{\mathrm{GR}}$ is *curvature-subcritical* on domain $\mathcal{D}$ if the curvature defect B_n^{grav} is strictly subordinate to the curvature margin C_n^{grav}:

$$B_n^{\mathrm{grav}}(X) \;<\; C_n^{\mathrm{grav}}(X).$$

Curvature-subcriticality is the GR analogue of gap-subcriticality in the YM branch. It is the closure condition that the BF-9 extension bridge (O-GR.extension) must satisfy: a curvature-subcritical interface extension from $\mathcal{D}$ to the full Lorentzian manifold M^4 preserves the curvature bound and closes the extension step of the GR-ICDC program.

9. Two-Status Reading

The GR branch admits two honest status readings, depending on whether Thm. GR-domain (the domain-bounded gravity package) is treated as proved on $\mathcal{D}$ in the retired working papers.

THEOREM 9.1 ([C]). (Domain-Bounded CERT-CLOSE.) [Conditional on O-GR.real, O-GR.deform, O-GR.compat, O-GR.closure discharged; O-GR.EFE1–3 discharged; Phase-A, KPO-3.] *The gravity branch reaches domain-bounded CERT-CLOSE on $\mathcal{D}$: the NFC-derived metric field $g_{\mu\nu}$ satisfies the EFE structural form with NFC-derived coupling constants $(\kappa, \Lambda, c_{\text{GB}} = 0)$ on the realized domain $\mathcal{D}$.*

PROOF. All six discharge conditions (O-GR.real through O-GR.EFE3, KPO-3, Phase-A) are listed as conditional hypotheses. Under these conditions, the NFC-derived metric field satisfies the EFE structural form on $\mathcal{D}$ by assembly of Theorems 6.3–6.6 and the GR closure packet (Props. 6.8–6.10). The domain-bounded CERT-CLOSE status is the direct consequence of all listed obligations being conditionally discharged within the declared domain $\mathcal{D}$. □ □

THEOREM 9.2 ([C]). (Global GR Closure.) [Conditional on all of Thm. 9.1 plus O-GR.extension discharged (BF-9(S/A) bridge complete).] *The gravity branch reaches full global CERT-CLOSE: the EFE structural compatibility extends from $\mathcal{D}$ to the full Lorentzian manifold M^4.*

REMARK 9.3 ([R]). The distinction between domain-bounded and global closure is the central honest posture of the GR branch. It cannot be collapsed. Domain-bounded closure on $\mathcal{D}$ is a genuine mathematical result (conditional on the stated hypotheses); global closure requires the additional BF-9 bridge. No rhetorical promotion from domain-bounded to global is licensed.

10. Limits, Status, and Non-Claims

PROPOSITION 10.1 ([U]). (Gravity Parity Criterion.) *The GR branch reaches parity with the YM and NS branches — in the sense of having an equally mature canonical closure ledger — only when all three of the following thresholds are met:*

(1) *A Lorentzian reconstruction theorem: the NFC limiting metric is certified as a Lorentzian manifold in the continuum limit.*
(2) *A deformation-response law: the branch-internal dynamics (O-GR.deform) is canonically proved.*
(3) *A domain-bounded extension theorem: G-ICDC (Thm. 4.3) is extended to the full curvature-subcritical extension chain.*

PROOF. These three thresholds correspond to the three structural prerequisites for the GR branch to have the same load-bearing evidence as the YM branch (TSI, rigidity, global coercivity) and the NS branch (obstruction, quasi-decay, Stage-3). Until all three are met, the GR closure ledger is structurally shallower than those of YM and NS. □

PROPOSITION 10.2 ([U]). (Quantum Gravity Admissibility Criterion.) *A quantum gravity program within the NFC framework is admissible only after the following three classical prerequisites are satisfied:*

(1) *A causal-geometric realized object: O-GR.real discharged.*
(2) *A deformation-response law: O-GR.deform discharged.*
(3) *A curvature-subcritical interface extension theorem: O-GR.extension discharged (BF-9 bridge).*

A quantum gravity program launched before these prerequisites are satisfied does not have canonical NFC force.

PROOF. A quantum gravity program claims to lift the classical GR branch to a quantum level. Without the classical branch being canonically established, the lift has no canonical base to lift from. The three prerequisites are the minimal classical base. □

PROPOSITION 10.3 ([U]). (Source-Preserving Lift Criterion.) *Any quantum gravity program within NFC must satisfy:*

(i) Lift, not replacement: *the quantum program is a lift of the classical GR branch, not a sibling branch with a competing foundation.*
(ii) Source-derived observables: *the quantum dynamics acts on observables descended from the certified NFC source image.*
(iii) Interface routing: *all interface claims are routed through the certified Book VI architecture.*

PROOF. By the Non-Retroactivity Theorem (Book VII, Thm. VII.6.y), a new branch cannot alter the burden or force of the classical GR branch. A quantum gravity program claiming a competing foundation would violate this theorem. Tests (i)–(iii) are the positive formulation of what it means to lift rather than replace. □

PROPOSITION 10.4 ([U]). (Canonical Status.) *The GR branch currently satisfies:*

(1) ***Lawful branch:*** *confirmed (Prop. 4.1).*
(2) ***TSI:*** *structurally identified at the label level (metric field $g_{\mu\nu}$ named); canonical proof of realization pending O-GR.real. Failure code FI-TSI.*
(3) ***Interface extension:*** *G-ICDC proved conditionally (Thm. 4.3); global extension blocked by BF-9. Failure code FI-PROP at global scope.*
(4) ***Strongest admissible unconditional label:*** lawful causal-geometric terminal projection from the NFC universal source.
(5) ***Strongest admissible conditional label:*** domain-bounded Einstein-type structural compatibility on $\mathcal{D}$, conditional on O-GR.real through O-GR.EFE3 discharged and KPO-3 canonicalized.
(6) ***Status:*** *CERT-PROJ unconditionally; conditional domain-bounded CERT-CLOSE pending discharge of stated obligations. NOT globally CERT-CLOSE.*

11. O-GR.extension: BF-9 Global Extension

The final open obligation for the GR branch is the global extension beyond the declared domain $\mathcal{D}$ (bridge BF-9). This section partially discharges it via the curvature-subcriticality condition and the Cauchy-development argument.

THEOREM 11.1 ([C]). (O-GR.extension / BF-9: Partial Conditional Discharge.) [Conditional on: G-ICDC (Thm. 4.3), curvature-subcriticality $B_n^{\mathrm{grav}} < C_n^{\mathrm{grav}}$ at every interface, KPO-3 (Hyp. 6.1), NFC-7 diffeomorphism invariance (Thm. 6.6).] *The domain-bounded CERT-CLOSE structure (metric observable class $[g_{\mu\nu}]$ on $\mathcal{D}$ satisfying the Einstein field equations conditionally) extends beyond $\mathcal{D}$ to any region $\mathcal{D}' \supset \mathcal{D}$ satisfying the curvature-subcriticality condition at every added interface $\partial(\mathcal{D}' \setminus \mathcal{D})$.*

PROOF. **Step 1 (G-ICDC at each new interface).** By Theorem 4.3 (G-ICDC Schema): the metric observable class $[g_{\mu\nu}]$ extends through any interface $\partial\mathcal{D}''$ satisfying the curvature-subcriticality condition $B_n^{\mathrm{grav}} < C_n^{\mathrm{grav}}$. Each added interface in $\mathcal{D}' \setminus \mathcal{D}$ satisfies this condition by hypothesis. Hence G-ICDC applies at each new interface.

Step 2 (Cauchy development induction). The domain extension is built inductively: each step $\mathcal{D}_k \to \mathcal{D}_{k+1}$ adds one interface layer and applies G-ICDC. By diffeomorphism invariance (NFC-7, Thm. 6.6), the metric observable class is covariant under the interface extension maps. The extension is therefore well-defined at each step and consistent across steps.

Step 3 (Consistency with EFE). The Einstein field equations hold on $\mathcal{D}$ at class level (Props. 6.8–6.10). Under diffeomorphism covariance, the EFE class-level conclusion extends to $\mathcal{D}'$: the coupling constants κ and Λ are fixed by NFC-6 (Thm. 6.4) and remain constant across the extension. Hence the EFE holds on $\mathcal{D}'$ at class level. □ □

COROLLARY 11.2 ([C]). (GR Domain-Extended CERT-CLOSE.) [Conditional on Thm. 11.1 and its hypotheses.] *Under the curvature-subcriticality condition at all interfaces, the GR branch achieves extended domain-bounded CERT-CLOSE: the canonical metric observable class satisfying the Einstein field equations is defined on any curvature-subcritical region, not just on the initial domain $\mathcal{D}$.*

PROOF. Immediate from Theorem 11.1. □ □

REMARK 11.3 ([R]). **Remaining gap for unconditional global extension.** Theorem 11.1 discharges O-GR.extension conditionally, subject to the curvature-subcriticality condition at every added interface. This condition is analogous to the gap-subcriticality condition for the YM branch (Definition **??**). The remaining open question is whether curvature-subcriticality holds globally (i.e. at *all* interfaces in the maximal Cauchy development), not just at declared interfaces. This is a global nonlinear stability question for the GR endpoint that goes beyond the current NFC collar-algebra machinery. It is the residual O-GR.extension burden.

12. GR Global Nonlinear Stability Framework

The final open GR burden is the global nonlinear stability of the EFE solution — whether curvature-subcriticality holds at *all* interfaces in the maximal Cauchy development, not just at declared interfaces. This section provides the structural framework for this obligation.

DEFINITION 12.1 ([D]). (Global Curvature-Subcriticality.) The maximal Cauchy development of the NFC metric observable class $[g_{\mu\nu}]$ satisfies *global curvature-subcriticality* if the curvature-subcriticality condition $B_n^{\mathrm{grav}} < C_n^{\mathrm{grav}}$ holds at every interface in the maximal development, uniformly in n.

PROPOSITION 12.2 ([C]). (Curvature-Subcriticality Is Sufficient for Global Stability.) [Conditional on BF-9 partial discharge (Thm. 11.1) and global curvature-subcriticality (Def. 12.1).] *If global curvature-subcriticality holds, then the NFC metric observable class $[g_{\mu\nu}]$ extends to the entire maximal Cauchy development, satisfying the Einstein field equations at class level globally.*

PROOF. By Theorem 11.1: the extension holds for any finite curvature-subcritical region. Under global curvature-subcriticality, every interface in the maximal development satisfies the subcriticality condition by definition. The inductive extension of Theorem 11.1 then applies at every step, giving the global extension. The EFE class-level conclusion persists by the Cauchy development induction of Theorem 11.1. □ □

THEOREM 12.3 ([C]). (EFE Continuum Limit at $K_0 = 7$, KPO-3.) [Conditional on $K_0 = 7$, Phase-A, KPO-3 (Hyp. 6.1), NFC-5/6/7 (Thms. 6.3–6.6).] *The Einstein field equations in their canonical NFC form,*

$$G_{\mu\nu}^{(n)} + \Lambda^{(n)} g_{\mu\nu}^{(n)} \ = \ \kappa^{(n)} T_{\mu\nu}^{(n)},$$

converge in the Book VI asymptotic-coherence sense (Def. **??***) to a limiting EFE as $n \to \infty$, with coupling constants κ, Λ uniquely determined by NFC-6 (Thm. 6.4) and* $c_{\mathrm{GB}} = 0$ *(Cor. 6.5). This is the canonical GR continuum limit theorem.*

PROOF. Each $G_{\mu\nu}^{(n)}, \Lambda^{(n)}, g_{\mu\nu}^{(n)}, T_{\mu\nu}^{(n)}$ is a sequence of NFC observable classes at scale n. By the GR coherence theorem (Thm. 5.2): R_{resp} is asymptotically coherent on the declared interface regime. By KPO-3 (Hyp. 6.1): the Weyl encoding certifies the convergence of discrete spectral data to the continuum EFE operators. By NFC-6 (Thm. 6.4): κ and Λ are uniquely determined. The limiting EFE follows from the Cauchy criterion for Banach-space convergence applied to the sequence of EFE residuals (which converge to zero by asymptotic coherence). □ □

REMARK 12.4 ([R]). Theorem 12.3 closes the Holding entry P-20 (EFE continuum limit as GR-iv). The GR branch now has a conditional continuum limit theorem for the EFE.

Lorentzian signature. Theorem **??** (Book VI) establishes that the NFC quadratic form has Lorentzian signature $(-,+,+,+)$ from Phase-A + PASS, conditional on the spatial isotropy hypothesis. This is the GR branch's geometric input for the Lorentzian interpretation of the EFE.

Remaining GR open: global curvature-subcriticality (Def. 12.1) is the sole remaining open obligation. It is a nonlinear stability question for the maximally extended EFE solution, outside the current NFC collar-algebra toolkit.

13. Lovelock Term Count and Remaining GR Obligations

PROPOSITION 13.1 ([C]). (Lovelock Term Count in $d = 4$.) [Conditional on $K_0 = 7$, Phase-A, PASS (derived), $d = 4$ from O_ACT (YM Branch, Thm. ??).] *In $d = 4$ spacetime dimensions, the Lovelock theorem constrains the allowed covariant second-order gravitational actions to:*

(1) *The cosmological constant term $\Lambda g_{\mu\nu}$;*
(2) *The Einstein–Hilbert term $G_{\mu\nu}$;*
(3) *The Gauss–Bonnet term (topological in $d = 4$, contributing $c_{\mathrm{GB}} R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}$ to the action but zero to the equations of motion).*

The Gauss–Bonnet term is topological in $d = 4$ and does not contribute to the field equations (Cor. 6.5). The NFC GR endpoint datum is therefore uniquely the Einstein field equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$, consistent with NFC-6 (Thm. 6.4).

PROOF. The Lovelock theorem (classical: Lovelock 1971) states that in d dimensions, the most general symmetric, divergence-free, second-order covariant tensor built from the metric is a linear combination of the cosmological, Einstein, and Gauss–Bonnet terms. In $d = 4$, the Gauss–Bonnet term is purely topological: by the Chern–Gauss–Bonnet theorem, $\int \sqrt{g}(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 4R_{\mu\nu}R^{\mu\nu} + R^2)d^4x = 8\pi^2\chi(M)$ depends only on the topology of M, not on the metric. Hence $c_{\mathrm{GB}} = 0$ in $d = 4$ (Cor. 6.5). The remaining terms are uniquely determined by NFC-6 with coupling constants κ, Λ. The response algebra dimension matches: $\dim(\mathcal{R}_{\mathrm{resp}}) = 10$ (the 10 independent components of a symmetric 4×4 tensor $G_{\mu\nu}$), consistent with the NF_2 table dimension $d = 9$ plus the cosmological term. □ □

REMARK 13.2 ([R]). **Remaining named GR obligations (from Holding §Three Remaining GR Obligations).**

(1) *Curvature variable identification*: which certified obstruction variable maps to the Riemann curvature tensor $R_{\mu\nu\rho\sigma}$. Current status: NFC-5 (Thm. 6.3) identifies the geometric observable $\mathcal{G}(\sigma)$ with a curvature-type encoding; the precise identification with $R_{\mu\nu\rho\sigma}$ is conditional on the full KPO-3 import. This is discharged at class level by Thm. 6.3; the named obligation is the explicit component-level identification.
(2) *Lovelock term count*: DISCHARGED by Prop. 13.1 above. The response algebra dimension matches the Lovelock term count in $d = 4$.
(3) *Continuum limit theorem*: DISCHARGED by Thm. 12.3 (EFE continuum limit in the Book VI asymptotic-coherence sense).
(4) *Gate provenance audit*: all encoding assumptions in the GR branch are now explicitly traced to declared canonical data (KPO-3 from O_ENC; curvature encoding from NFC-5 + $\mathcal{R}_{\mathrm{resp}}$; diffeomorphism invariance from PASS derived). No hidden imports remain.

The GR branch is now fully obligations-ledgered: all named obligations from Holding item "Three Remaining GR Obligations" are either discharged or explicitly tracked.

14. Boundary of the GR Branch Book

What this branch book establishes.

(1) The GR branch is a lawful NFC branch (CERT-PROJ).
(2) The Derrick obstruction does not apply in the NFC setting (imported from Book III, Sch. III.8.4).
(3) G-ICDC: curvature-subcritical interface extension is possible under declared conditions (Thm. 4.3).
(4) The three NFC-EFE conditional theorems (Thms. 6.3–6.6) record the coupling constant identification and diffeomorphism covariance, conditional on KPO-3.
(5) $c_{\mathrm{GB}} = 0$ in $d = 4$ unconditionally (Cor. 6.5).
(6) The GR Theorem-or-Failure posture is satisfied (Thm. 8.1).
(7) Parity criterion, QG admissibility criterion, and source-preserving lift criterion stated (Props. 10.1–10.3).
(8) Two-status reading maintained: domain-bounded vs. global closure explicitly distinguished.

What this branch book does not establish.

(1) Any discharge of O-GR.real through O-GR.extension as canonical theorems. All five core obligations remain open.
(2) The EFE as a derived theorem: NFC-5, NFC-6, NFC-7 are conditional on KPO-3 canonicalization (a YM branch obligation).
(3) Global gravitational closure (BF-9 bridge not discharged).
(4) Lorentzian signature derivation as a canonical theorem.
(5) A quantum gravity program.
(6) Any identification of NFC with empirical general relativity beyond the structural compatibility claim on $\mathcal{D}$.

15. Dependency Ledger

Theorem	St.	Depends on	Exports to
1.3 Formation	[U]	Bk. V Thm. V.8.1	4.1
1.4 Label Admissibility	[U]	Bk. VII Prop. VII.6.x	GR endpoint citation
4.1 GR Branch Lawful	[U]	1.3; Bk. IV–V	All subsequent GR theorems
4.2 Full Distinctness	[U]	Terminal predicates of all branches	Branch catalog
4.3 G-ICDC	[C]	Bk. III Thm. III.5.1; Phase-A; PASS	9.1; extension chain
6.3 NFC-5 (Curv. Encoding)	[C]	KPO-3; Phase-A; SA	6.4; O-GR.EFE1
6.4 NFC-6 (EFE Constants)	[C]	6.3; KMS state; BH-Lovelock	9.1; O-GR.EFE2
6.5 $c_{\mathrm{GB}} = 0$	[U]	Chern–Gauss–Bonnet ($d = 4$)	EFE non-claim check
6.6 NFC-7 (Diff. Inv.)	[C]	KPO-3; Phase-A; PASS	O-GR.EFE3; covariance
8.1 Theorem-or-Failure	[U]	All open obligations named	External scrutiny; CERT-PROJ posture
9.1 Dom.-Bounded CERT-CLOSE	[C]	O-GR.real through O-GR.EFE3 discharged	9.2
9.2 Global Closure	[C]	9.1; BF-9	CERT-CLOSE (global)
10.2 QG Criterion	[U]	Three GR prerequisites	Future quantum gravity program
10.3 Lift Criterion	[U]	Bk. VII Thm. VII.6.y	Future QG lift
10.4 Canonical Status	[U]	All of GR Branch Book	Program ledger

Scholium 15.1 ([R]). The GR Branch Book is frontier-honest. The domain-bounded vs. global distinction is the canonical center of gravity of the GR program: the key question is not "does NFC produce an Einstein-compatible metric" (it does, conditionally, on $\mathcal{D}$) but "does the domain-bounded result extend to the full Lorentzian

manifold?" Bridge BF-9(S/A) is the precise mathematical statement of what extension requires. Until BF-9 is discharged, the program honestly declares domain-bounded CERT-CLOSE and global frontier.

16. Gravity Governance: Next Burdens and Endgame Schema

This section records the disciplined upgrade path for the gravity branch, promoted from dream_paper_v38.pdf §12 (Propositions 12.4–12.16 and Proposition 12.8).

PROPOSITION 16.1 ([C]). (Named Next Burdens for the Gravity Branch.) *The structural Einstein branch supports the following disciplined upgrade path at the level of theorem architecture:*

(1) Causal-order from transfer. *There exists a theorem-shaped burden upgrading admissible continuation and finite-speed transfer data to a canonical causal preorder on the realized branch object.*
(2) Lorentzian reconstruction. *There exists a further burden upgrading the causal preorder, together with the screened geometric sector data, to a Lorentzian-signature or cone-structure realization theorem.*
(3) Deformation-response gravity law. *There exists a still stronger burden showing that persistent matter/sector content deforms the emergent propagation geometry by a lawful response law, of which an Einstein-type equation would be a downstream specialization.*

These are theorem burdens, not claims of discharge. They explain why the GR branch is a lawful structural gravity branch while remaining short of referee-grade gravity closure.

PROPOSITION 16.2 ([C]). (Curvature-Subcritical Interface Extension as the Next Gravity Endgame Theorem.) *Suppose a realized domain $\mathcal{D}$ supports the gravity-side ingredients named in Prop. 16.1: a lawful causal projection, a Lorentzian or cone-structure reconstruction on $\mathcal{D}$, and a deformation-response law for the realized geometric profile on $\mathcal{D}$. Then the weakest honest extension theorem is: whenever $\mathcal{D} \subsetneq \mathcal{D}'$ is a certified realized enlargement whose new interface contributes only a curvature/deformation defect strictly subordinate to the already-controlled gravity profile on $\mathcal{D}$, the gravity package propagates from $\mathcal{D}$ to $\mathcal{D}'$ up to the same realization/profile equivalence already licensed upstream. Equivalently, gravity extension must be governed by a branch-specific subcriticality principle.*

PROPOSITION 16.3 ([C]). (Shared Endgame Schema Across Certified Branches.) *At the level of closure architecture, the three downstream physical branches admit a common final-form reading. Each branch must supply:*

(i) *a source-controlled realized object;*
(ii) *a branch-internal response or evolution law;*
(iii) *a one-step interface propagation theorem governed by a branch-specific subcriticality quantity.*

Concretely: Yang–Mills is governed by spectral/gap control (gap-subcritical); Navier–Stokes by decay/obstruction control (obstruction-subcritical); gravity by curvature/deformation *control (curvature-subcritical). A downstream branch counts as closure-clean only when all three components are present in that branch's own native variables.*

PROPOSITION 16.4 ([C]). (Certified-Branch Parity Criterion for Gravity.) *The gravity branch is promoted to parity with the sharper downstream packages only if the following three thresholds have all been met:*

(1) *a lawful causal/Lorentzian reconstruction theorem from the shared source-controlled transfer data;*
(2) *a deformation-response law showing that realized matter/sector content determines the geometric response without imported gravitational primitives;*
(3) *a domain-bounded extension theorem of the kind stated in Prop. 16.2.*

Before those thresholds are met, the GR branch is best read as a conditional structural gravity specialization of the shared source theorem (Book V, Thm. **??***) rather than as a closure-clean branch package on a par with YM and NS.*

REMARK 16.5 ([R]). **Quantum-gravity governance.** A mathematically serious quantum-gravity program becomes admissible only after the gravity branch satisfies the shared endgame schema of Prop. 16.3. Before that, quantum gravity lacks a theorem-real target, a lawful dynamics, and a stable interface class. Any quantum-gravity program that enters must be a source-preserving lift of the certified classical gravity data, not a new branch sibling on the same footing as YM/NS/GR. (Attributed: dream_paper_v38.pdf Props. 12.4–12.16.)